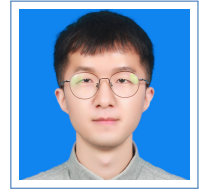


# Yunqiang Pei

Simon

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🌐 [simon-pei.github.io/AcademicHomePage/](https://simon-pei.github.io/AcademicHomePage/)  
🔍 [google scholar](#)



## Research Interest

Augmented Reality × AI, Human-Agent Alignment, Physiological Computing

## Professional Experience

Mar. 2025 – **KAIST**, *South Korean*

Mar. 2026 ○ Visiting student researcher conducting research in the WIT Lab, under the supervision of Professor Ian Oakley

## Education

Sep. 2022 – **Ph.D. Candidate in Electronic Information**, *University of Electronic Science and Technology of China (UESTC)*, Chengdu, China

○ **GPA:** 3.84 / 4.0

○ School of Computer Science and Engineering

Sep. 2018 – **M.Sc. in Computer Science and Technology**, *Southwest University of Science and Technology*, Mianyang, China

○ **GPA:** 90.88 / 100.0 (**Ranked 1<sup>st</sup>** in the whole grade)

○ School of Computer Science and Technology

Sep. 2014 – **B.S. in Internet of Things Engineering**, *Sichuan University of Science and Engineering*, Zigong, China

○ **GPA:** 3.7 / 5.0 (**Ranked 1<sup>st</sup>** in Major)

○ School of Computer Science

## Employment

Jul. 2021 – **Software engineer**, *Thoughtworks*, *Chengdu, Sichuan, China*.

Nov. 2021

## Honors and Awards

Class of 2026 Outstanding Graduate Student (Projected Honor)

2025 Academic Rising Star Honor of UESTC

2024 **Best Paper Nomination**, ACM Multimedia (MM) 2024

2024 First-Class Scholarship

- 2023 Second Prize, National University Student Five Minute Research Presentation (English)
- 2023, 2025 Second-Class Scholarship (Awarded twice)
- 2022 Freshman Scholarship
- 2021 **Outstanding Master's Thesis**
- 2019 **Best Paper Honorable Mention Award**, ICVRV 2019
- 2015 – 2017 **National Scholarship** (Awarded for **two consecutive years**)
- 2018 Sichuan Province Outstanding Graduate
- 2014 – 2017 First-Class Scholarship (Awarded for three consecutive years)

## Publications

Quick summary since 2019: First Author(8), Awarded Paper(1), Best Paper Nomination (1), MM(3), VR (poster × 2), UIST(1), AAAI(1), ICVRV(1), PRAI(1), SCIENCE CHINA Information Sciences(1), Journal of System Simulation(1), Journal of Graphics(2)

- C1 **Yunqiang Pei**, Hongrong Yang, Kaiyue Zhang, Guoqing Wang\*, Peng Wang, Chaoning Zhang, Yang Yang, Heng Tao Shen. InteractGuide: LLM-Enhanced Multimodal Reasoning for User-Centric Interaction Recommendations in AR-HRI Authoring. *ACM Multimedia* (MM Oral Paper). 2025
- C2 **Yunqiang Pei**, Renming Huang, Mingfeng Zha, Guoqing Wang\*, Peng Wang, Kang Qiao, Yang Yang, Heng Tao Shen. AttentionAR: AR Adaptation and Warning for Real-World Safety via Attention Modeling and MLLM Reasoning. *ACM Symposium on User Interface Software and Technology* (UIST Paper). 2025
- C3 **Yunqiang Pei**, Kaiyue Zhang, Hongrong Yang, Yong Tao, Qihang Tang, Jialei Tang, Guoqing Wang\*, Zhitao Liu, Ning Xie, Yang Yang, Heng Tao Shen. Improving Interaction Comfort in Authoring Tasks in AR-HRI through Dynamic Dual-Layer Interaction Adjustment. *ACM Multimedia* (**Best Paper Nomination**). 2024
- C4 **Yunqiang Pei**, Qihang Tang, Jialei Tang, Mingfeng Zha, Dongyu Xie, Guoqing Wang\*, Zhitao Liu, Ning Xie, Yang Yang, Heng Tao Shen. Emotion Recognition in HMDs: A Multi-task Approach Using Physiological Signals and Occluded Faces. *ACM Multimedia* (MM Paper). 2024
- C5 **Yunqiang Pei**, Kaiyue Zhang, Ziyang Lu, Mingfeng Zha, Guoqing Wang\*, Zhitao Liu, Ning Xie, Yang Yang, Heng Tao Shen. Toward Optimized AR-based Human-Robot Interaction Ergonomics: Modeling and Predicting Interaction Comfort. *IEEE Conference on Virtual Reality and 3D User Interfaces* (VR Poster). 2024
- C6 Ziyang Lu, **Yunqiang Pei**, Guoqing Wang\*, Peiwei Li, Yang Yang, Yinjie Lei, Hengtao Shen. ScanERU: Interactive 3D Visual Grounding Based on Embodied Reference Understanding. *The 38th Annual AAAI Conference on Artificial Intelligence*. 2024
- C7 **Yunqiang Pei**, Renming Huang, Guoqing Wang\*, Yang Yang, Heng Tao Shen. Multimodal Apology: Using WebXR to Repair Trust with Virtual Companion. *IEEE Conference on Virtual Reality and 3D User Interfaces* (VR Poster). 2023

- C8 Shanshan Xiang, Wensheng Tang, **Yunqiang Pei**, Yadong Wu\*, Yonghui Chen, Fupan Wang, Zhiwei Hou. IV-WA: An Information Visual Tool with WebAR. *International Conference on Pattern Recognition and Artificial Intelligence (PRAI)*. 2021
- C9 **Yunqiang Pei**, Yadong Wu\*, Song Wang, Fupan Wang, Hongyu Jiang, Shijian Xu, Jinquan Zhou. Wa vis: A Web-based Augmented Reality Text Data Visual Analysis Tool. *International conference on virtual reality and visualization (ICVRV, Best Paper Honorable Mention Award \*)*. 2019
- J1 Guoqing Wang (Advisor)\*, **Yunqiang Pei**, Yang Yang, Xing Xu, Zheng Wang, Hengtao Shen. A Review and Outlook of Shared Multi-Modal Trustworthy Human-Robot Interaction Research, *SCIENCE CHINA Information Sciences*. 2023
- J2 Chunhong Liu, Song Wang\*, Fupan Wang, Wensheng Tang, **Yunqiang Pei**, Dongsheng Tian, Yadong Wu. AR-assisted Sign Language Letter Recognition Method Based on Improved MobileNet Network, *Journal of System Simulation*. 2023
- J3 Nanshan Liu, **Yunqiang Pei**, Hao Jiang, Yongguo Han, Yadong Wu\*, Fupan Wang, Siheng Yi. WebAR garbage classification information visualization method based on VD-MobileNet network, *Journal of Graphics*. 2022
- J4 **Yunqiang Pei**, Yadong Wu\*, Fupan Wang, Xiaorong Zhang, Hongyu Jiang, Shijian Xu, Wensheng Tang. IV LKWA: an information visual analysis tool with advanced L-K optical flow based WebAR, *Journal of Graphics*. 2020

## Peer Review Service

AAAI, UIST, CHI, CSCW, ISMAR, VRST, UbiComp/ISWC, IUI, MM, DIS, IDC, IMX

## Winter and Summer Vacation Study Program

Jan. 2023 – I participated in the Machine Learning Winter Programme organised by **Girton College, Cambridge** (Online)  
Feb. 2023

## Conference Volunteering Experience

Jul. 2019 Student Volunteer at **ChinaVIS 2019**, Chengdu, China.  
Apr. 2025 Student Volunteer at **ACM CHI 2025**, Yokohama, Japan.  
Oct. 2025 Student Volunteer at **ISMAR 2025**, Daejeon, South Korea.

## Hardware Experience (Software Utilization)

HMD & Glass Hololens 2, XReal Air 2 Ultra, 1S, Beam Pro, Eye  
PPG & GSR Shimmer3R GSR+  
EEG & IMU Brainco Oxyzen, OpenBCI Ganglion

## Invited Talks

Mar. 2025 Enhancing XR Interactions: Comfort, Emotion, and Safety — Lunch Seminar, WIT Lab, KAIST

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## Selected Press Coverage

Nov. 2025 UESTC Takes Me to See the World · Exploring the Future of Human-AI Symbiosis at the Intersection of AR and AI

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## Languages and Levels

Chinese Proficient  
English Advanced  
Korean Intermediate